Protocol: I8B-MC-ITSA(d)

# A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 Compared to Humalog in Children, Adolescents, and Adults With Type 1 Diabetes Mellitus

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# Protocol I8B-MC-ITSA(d)
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EudraCT Number: 2017-003220-78

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LY900014

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# 1. Protocol Synopsis

**Title of Study:**

A study to evaluate the pharmacokinetics and glucodynamics of LY900014 compared to Humalog in children, adolescents, and adults with type 1 diabetes mellitus

**Rationale:**

A prandial insulin with faster-on and/or faster-off characteristics might reduce glycaemic excursions and the incidence of postprandial hypoglycaemia compared to currently available fast-acting insulin analogues. LY900014 is an ultra-rapid-acting insulin lispro formulation with increased early absorption compared to commercially available insulin lispro formulation (Humalog®; Eli Lilly and Company). LY900014 aims to mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions.

The aim of the study is to evaluate insulin lispro pharmacokinetics (PK), glucodynamics (GD), safety, and tolerability of LY900014 in comparison with Humalog following a single subcutaneous (SC) dose administration immediately prior to a standardised liquid meal in paediatric and adult patients with type 1 diabetes mellitus (T1DM). This will be the first study to evaluate LY900014 in paediatric patients with T1DM.

**Objectives/Endpoints:**

| Objectives | Endpoints |
|---|---|
| **Primary** | |
| 1. Part A: To evaluate the PK of insulin lispro following a single SC dose administered through injection of LY900014 compared to Humalog in children, adolescents, and adults with T1DM | 1. Early 50% $t_{max}$ and AUC(0-30min) |
| 2. Part B: To evaluate the PK of insulin lispro following a single SC bolus dose administered through CSII of LY900014 compared to Humalog in children, adolescents, and adults with T1DM | 2. Early 50% $t_{max}$ and AUC(0-30min) |
| **Secondary** | |
| 1. Part A and Part B: To evaluate the difference in GD response to LY900014 and Humalog administered through SC bolus injection or CSII, as assessed using the liquid MMTT in children, adolescents, and adults with T1DM | 1. ΔAUC(0-2h), ΔAUC(0-5h) |
| 2. Part A and Part B: To evaluate the tolerability of LY900014 following SC injection or CSII in children, adolescents, and adults with T1DM | 2. AEs including injection/catheter insertion site reactions/pain, hypoglycaemia, and anti-insulin lispro antibodies |

Abbreviations: AE = adverse event; AUC = area under the concentration versus time curve; AUC(0-30min) = AUC from time zero to 30 minutes; CSII = continuous subcutaneous insulin infusion; ΔAUC = area under the baseline subtracted glucose concentration versus time curve; ΔAUC(0-2h) = ΔAUC from time 0 to 2 hours postmeal; ΔAUC(0-5h) = ΔAUC from time 0 to 5 hours postmeal; early 50% $t_{max}$ = time to early half-maximal drug concentration; GD = glucodynamics; MMTT = mixed meal tolerance test; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus.

**Summary of Study Design:**

Study I8B-MC-ITSA is a randomised, patient- and investigator-blind, 2-part study, with each part containing a 2-period crossover assessment in children (age 6 to <12 years), adolescents (age 12 to <18 years), and adults (age 18 to <65 years) with T1DM to evaluate the PK, GD, and tolerability of LY900014 compared to Humalog given as a bolus either as a SC injection or via a continuous subcutaneous insulin infusion (CSII) pump.

**Treatment Arms and Planned Duration for an Individual Patient:**
The study will be conducted in 2 parts, Part A and Part B. Each part consists of a 2-period crossover to compare LY900014 and Humalog administered prior to a mixed meal tolerance test (MMTT). All study and treatment procedures will be conducted during inpatient stays of approximately 2 days per period and require a minimum of 3 days and a maximum of 23 days to complete both periods. In each part, after completion of the last study period (Period 2), a 14-day safety follow-up is planned. Total duration of each part is expected to be up to 68 days. Patients are encouraged to participate in both Part A and Part B of the study; however, study participation is not contingent on patients' participation in both parts of the study.

Patients may be screened up to 28 days prior to Day -1 of Period 1 of Part A and Part B. Patients will be admitted to the study site the day before the MMTT day each period and will remain until all assessments are completed on Day 1. On Day 1 in Period 1 for Part A and on Day -1 in Period 1 for Part B, patients will be randomised to 1 of 2 treatment sequences. In Part A, each patient will receive a single SC injection of both LY900014 and Humalog immediately prior to an MMTT assessment in a crossover design. Initiation of Part B of the study may begin pending the review of the PK, GD, safety, and tolerability data from Part A and ongoing or planned adult CSII Phase 1b studies. In Part B, patients will receive a single SC bolus dose of both LY900014 and Humalog via CSII pump immediately prior to an MMTT assessment in a crossover design. Patients will undergo a follow-up visit at least 14 days after the last dose of the study drug in each part.

**Treatment:**

Part A: Each patient will receive a single SC bolus injection of LY900014 and Humalog.

Part B: Each patient will receive a single SC bolus of LY900014 and Humalog via CSII pump.

**Number of Patients:**
Approximately 45 patients may be enrolled into the study so that approximately 36 patients (12 patients in each age group [6 to <12 years, 12 to <18 years, and 18 to <65 years] based on the age at initial screening) complete both inpatient periods in both parts of the study.

Patients participating in Part A may continue participation in Part B. If patients from Part A decline participation in Part B or drop out from Part A, additional patients will be enrolled to ensure approximately 36 patients (12 patients in each age group) complete both periods in Part B of the study.

**Statistical Analysis:**
The analyses for Part A and Part B will be conducted separately using similar statistical methods. Primary statistical analyses will be conducted on the set of patients who complete both treatment periods. Supportive analyses may be done on the key parameters for the patients who complete at least 1 treatment period. Safety analyses will be conducted for the set of patients receiving at least 1 dose of the study drug, whether or not they completed all protocol requirements.

**Safety:** All study drug- and protocol procedure-related adverse events will be listed, and if the frequency of events allows, safety data (including hypoglycaemic events) will be summarised using descriptive methodology. Safety parameters will be listed, and may be summarised using standard descriptive statistics.

**Pharmacokinetic:** Log-transformed PK parameters for insulin lispro will be evaluated to estimate least-squares geometric means, ratios of geometric means between LY900014 and Humalog, and their corresponding 95% CIs of the ratios using the mixed-effect model that includes treatment, sequence, period, age group, and treatment by age group interaction as fixed effects and patient within sequence as a random effect.

The same model without log transformation will be used for the analysis of the PK time parameters (time to early half-maximal drug concentration [early 50% $t_{max}$], time to late half-maximal drug concentration [late 50% $t_{max}$], time of maximum observed drug concentration ($t_{max}$). Least-squares means (LSmeans), treatment differences in LSmeans, and the corresponding 95% CIs for the treatment differences will be estimated from the model. The treatment ratios and 95% CIs for the ratios using the Fieller's theorem will be provided.

**Glucodynamic**: Data will be analysed for the patients during each MMTT. The change from baseline values (the average of -30, -15, and 0 minutes) represented as the 0-hour time point following the start of the MMTT for each patient will be calculated. The area under the baseline subtracted glucose concentration versus time curve (ΔAUC) from time 0 to 2 hours postmeal (ΔAUC[0-2h]) and ΔAUC from time 0 to 5 hours postmeal (ΔAUC[0-5h]) will be calculated. In addition, the change from baseline maximum glucose observed during the 5 hours postmeal and change from baseline 1 hour glucose and 2 hour glucose after the start of the meal will be calculated. Other partial ΔAUCs may be calculated as deemed appropriate.

Summary statistics (including number of patients, mean, standard deviation or standard error, minimum, and maximum) will be presented by treatment for each age group. The GD parameters on the original scale (not log transformed) will be analysed using the same model used for PK parameters. Least-squares means, treatment differences in LSmeans, and the corresponding 95% CIs for the treatment differences will be estimated from the model. The treatment ratios and 95% CIs for the ratios will be calculated using the Fieller's theorem.

# 2. Schedule of Activities

**Study Schedule Protocol I8B-MC-ITSA (Part A)**

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 101 | Telephone Visit | 1 and 2 | | 801 | |
| Day | Within 28 days Prior to Day -1 | ≥Day -10[c] | -1[d] | 1 | ≥14 days After Last Dose of Study Drug | |
| Informed Consent/Assent | X | | | | | Should be obtained before any study-related procedures are performed. |
| Inclusion/Exclusion Criteria | X | | | | | |
| Medical History | X | | | | | |
| Pre-existing Conditions | X | | | | | |
| Age | X | | | | | |
| Height | X | | | | | |
| Weight | X | | | X | X | |
| Physical Examination | X | | | Predose | X | Physical examination at screening. Thereafter, targeted medical assessment, as appropriate. |
| Admission to Study Site | | | X | | | |
| IWRS and Randomisation | | | | X | | Randomisation in Period 1 only. Age group for randomisation will be based on initial screening. |
| Previous Diabetes Therapy Record Prandial and Basal Insulin Dose | X | | | X* | | *Period 1 only. |
| Vital Signs (Pulse, Sitting BP) | X | | | Pre- and 2 hours postdose | X | |
| Eligibility Assessment | X | X | X* | | | *Period 1 only. |
| Hip and Waist Circumference | | | X | | | Applicable only to adult patients. |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 101 | Telephone Visit | 1 and 2 | | 801 | |
| Day | Within 28 days Prior to Day -1 | ≥Day -10[c] | -1[d] | 1 | ≥14 days After Last Dose of Study Drug | |
| Clinical Laboratory Tests | X | | | | X | See Appendix 2 for details. All patients of legal drinking age (according to local laws) will receive alcohol breath test on Day -1. |
| Glycated Haemoglobin (HbA1c) Assessment | X | | | | | |
| Urinalysis | X | | | | X | |
| 12-Lead ECG | X | | | | | |
| Pregnancy Test | X | | X | | X | Pregnancy tests will be performed for all females ≥12 years of age or <12 years of age with onset of menses. Serum pregnancy tests will be performed at screening, and urine pregnancy test on Day -1 of Periods 1 and 2 (or when clinically indicated), and at the follow-up or ED visit. Urine pregnancy tests at other visits accepted per local regulations. |
| Provide Evening Meal | | | X | | | |
| Begin Overnight Fast | | | X | | | Patients will be fasted (except for water) for at least 10 hours before each test meal. |
| Stop CSII | | | X | | | Prior to start of run-in. Applicable only to patients on |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 101 | Telephone Visit | 1 and 2 | | 801 | |
| Day | Within 28 days Prior to Day -1 | ≥Day -10[c] | -1[d] | 1 | ≥14 days After Last Dose of Study Drug | |
| | | | | | | CSII. |
| Run-in/Stabilisation of Glucose | | | X (following evening meal) | | | Approximately at 09:00 to 10:00 PM infusion of glucose (dextrose solution) or CCI insulin to target a blood glucose concentration of 135±25 mg/dL (7.5±1.4 mmol/L). Blood glucose concentrations will be monitored at a minimum of 30- to 60-minute intervals. |
| Study Drug Administration (Single SC injection) | | | | X | | Time of study drug administration = 0 hour. Study drug will be administered at approximately the same times on Day 1 of each study period. |
| Immunogenicity Sample | | | | Predose | X | |
| Injection-Site Evaluation | | | | 0, 60, and 240 minutes postdose | | Time 0: Assessments of injection-site local tolerability will occur immediately following the injection. |
| Directions for Preparing for PK and MMTT Visits | | X | | | | Patients will receive preparatory instructions as part of their pre-study telephone consultation (see Section 6.3, for details). |
| Mixed Meal Tolerance Test (MMTT) | | | | X | | A standard liquid meal will be administered at the MMTT time "0" (as soon as possible after |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 101 | Telephone Visit | 1 and 2 | | 801 | |
| Day | Within 28 days Prior to Day -1 | ≥Day -10[c] | -1[d] | 1 | ≥14 days After Last Dose of Study Drug | |
| | | | | | | administration of study drug) and should be consumed within 15 minutes. |
| Blood Glucose Sampling (MMTT) | | | | -30, -15, 0 (premeal), 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 135, 150, 165, 180, 195, 210, 225, 240, 300 minutes postmeal | | Time is referenced to meal start time, i.e., time 0. Time 0 is collected immediately prior to start of meal consumption. |
| Insulin Lispro PK Sampling | | | | 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 90, 120, 150, 180, 240, 300, 360, and 420 minutes postdose | | Sampling times are relative to the time of study drug treatment administration (t = 0) in each period |
| Transition to NPH Twice Daily Therapy 7-3 days Prior to Day -1 | | X | | | | Applicable only to patients using multiple daily injection therapy. Patient can return to pre-study basal insulin following completion of MMTT sampling period per investigator's decision and allowing for transition to NPH prior to next dosing. |
| Resume Prestudy CSII Rate | | | | X | | Applicable only to patients on CSII. CSII with the patient's prestudy insulin will be restarted after |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 101 | Telephone Visit | 1 and 2 | | 801 | |
| Day | Within 28 days Prior to Day -1 | ≥Day -10[c] | -1[d] | 1 | ≥14 days After Last Dose of Study Drug | |
| | | | | | | completion of assessment period (approximately 420 minutes postdose). |
| Adverse Events/Hypoglycaemia | X | X | X | X | X | |
| Concomitant Medications | X | X | X | X | X | |
| Patient Discharge from Study Site | | | | X | | If the next MMTT (Period 2) is scheduled on the next day, the patient may be discharged after the second MMTT (Period 2). |

Abbreviations: BP = blood pressure; CSII = continuous subcutaneous insulin infusion; ECG = electrocardiogram; ED = early discontinuation; IWRS = Interactive Web Response System; MMTT = mixed meal tolerance test; NPH = neutral protamine Hagedorn; PK = pharmacokinetics; SC = subcutaneous.

a Safety follow-up visit should occur at least 14 days after the last dose.

b Early discontinuation procedures will be performed at least 14 days after a patient stops the study after receiving study drug on Day 1 of Period 1 and before Day 1 of Period 2.

c This is a telephone visit.

d In case of dosing on consecutive days, the medical assessments to be conducted on Day -1 of Period 2 may be done on Day 1 of Period 1.

Note: The study site should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: pharmacokinetic samples including blood sampling for blood glucose and laboratory samples per protocol nominal times. Electrocardiograms and vital sign measurements should be scheduled before but as close as possible to the PK sampling times. Injection-site assessments can be done after PK sampling.

**Study Schedule Protocol I8B-MC-ITSA (Part B)**

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| Informed Consent/Assent | X[e] | | | | | Should be obtained before any study-related procedures are performed. |
| Inclusion/Exclusion Criteria | X | | | | | |
| Medical History | X | | | | | |
| Pre-existing Conditions | X | | | | | For patients continuing from Part A, a review will be performed to record any changes from the previous screening. |
| Age | X | | | | | Applicable only to newly enrolled patients. For patients participating from Part A, age at initial screening (Part A) will be used for reporting purposes. |
| Height | X | | | | | |
| Weight | X | | X | | X | |
| Physical Examination | X | | | Predose | X | Physical examination at screening. Thereafter, targeted medical assessment, as appropriate. |
| Admission to Study Site | | | X | | | |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| IWRS and Randomisation | | | X | | | Randomisation at Period 1 only. Age group for randomisation will be based on initial screening. |
| Previous Diabetes Therapy Record Prandial and Basal Insulin Dose | X | | | X* | | *Period 1 only. |
| Vital Signs (Pulse, Sitting BP) | X | | | Pre- and 2 hours postdose | X | |
| Hip and Waist Circumference | | | X | | | Applicable only to adult patients. |
| Eligibility Assessment | X | X | X* | | | *Period 1 only. |
| Clinical Laboratory Tests | X | | | | X | See Appendix 2 for details. All patients of legal drinking age (according to local laws) will receive alcohol breath test on Day -1. |
| Glycated Haemoglobin (HbA1c) Assessment | X | | | | | |
| Urinalysis | X | | | | X | |
| 12-Lead ECG | X | | | | | |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| Pregnancy Test | X | | X | | X | Pregnancy tests will be performed for all females ≥12 years of age or < 12 years of age with onset of menses. Serum pregnancy tests will be performed at screening, and urine pregnancy test on Day -1 of Periods 1 and 2 (or when clinically indicated), and at the follow-up or ED visit. Urine pregnancy tests at other visits accepted per local regulations. |
| Provide Evening Meal and Administer Bolus Insulin Infusion | | | X | | | Provide instructions for replacement of pump reservoir/tubing/infusion set containing IP to occur following evening meal. Prandial insulin to be patient's usual insulin. |
| Begin Overnight Fast | | | X | | | Patients will be fasted (except for water) for at least 10 hours before each test meal. |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| Reduce CSII basal rate | | | X | | | Prior to start of run-in. Applicable only to patients on CSII. Basal rate to be reduced to 0.1 U/hour. |
| Run-in/Stabilisation of Glucose | | | X (following evening meal) | | | Approximately at 09:00 to 10:00 PM infusion of glucose (dextrose solution) or CCI [redacted] insulin to target a blood glucose concentration of 135±25 mg/dL (7.5±1.4 mmol/L). Blood glucose concentrations will be monitored at a minimum of 30- to 60-minute intervals. |
| Study Drug Administration (Single Bolus via CSII) | | | | X | | Time of study drug administration = 0 hour. Study drug will be administered at approximately the same times on Day 1 of each study period. |
| Immunogenicity Sample | | | | Predose | X | |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| Catheter-Site Evaluation | | | | 0, 20, 40, 60, and 240 minutes postdose | | Time 0: Assessments will occur immediately following the study drug administration. |
| Directions for Preparing for PK and MMTT Visits | | X | | | | Patients will receive preparatory instructions as part of their pre-study telephone consultation (see Section 6.3, for details). |
| Mixed Meal Tolerance Test | | | | X | | A standard liquid meal will be administered at the MMTT time "0" (as soon as possible after administration of study drug) and should be consumed within 15 minutes. |
| Blood Glucose Sampling (MMTT) | | | | -30, -15, 0 (premeal), 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 135, 150, 165, 180, 195, 210, 225, 240, 300 minutes (postmeal) | | Time is referenced to meal start time. Time 0 is collected immediately prior to start of meal consumption. |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| Insulin Lispro PK Sampling | | | | 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 90, 120, 150, 180, 240, 300, 360, and 420 minutes postdose | | Sampling times are relative to the time of study drug treatment administration (t = 0) in each period. |
| Transition to NPH Twice Daily Therapy 7-3 days Prior to Day -1 | | X | | | | Applicable only to patients using multiple daily injection therapy. Patient can return to pre-study basal insulin following completion of MMTT sampling period per investigator's decision and allowing for transition to NPH prior to next dosing. |
| Resume Prestudy CSII Rate | | | | X | | Applicable only to patients on CSII. CSII with the patient's prestudy insulin will be restarted after completion of assessment period (approximately 420 minutes postdose). |
| Adverse Events/Hypoglycaemia | X | X | X | X | X | |
| Concomitant Medications | X | X | X | X | X | |

| Visits | Screening | | Study Periods (Periods 1 and 2) | | Follow-up[a]/Early Discontinuation[b] | Comments |
|---|---|---|---|---|---|---|
| | 102 | Telephone Visit | 3 and 4 | | 802 | |
| Day | Within 28 days Prior to Day -1 | ≥ Day -10[c] | -1[d] | 1 | ≥14 days After Discharge | |
| Patient Discharge from Study Site | | | | X | | If the next MMTT (Period 2) is scheduled on the next day, the patient may be discharged after the second MMTT (Period 2). |

Abbreviations: BP = blood pressure; CSII = continuous subcutaneous insulin infusion; ECG = electrocardiogram; ED = early discontinuation; IWRS = Interactive Web Response System; MMTT = mixed meal tolerance test; NPH = neutral protamine Hagedorn; PK = pharmacokinetics; SC = subcutaneous.

a Safety follow-up visit should occur at least 14 days after the completion of Period 2.

b Early discontinuation procedures will be performed at least 14 days after a patient stops the study after receiving study drug on Day 1 of Period 1 and before Day 1 of Period 2.

c This is a telephonic visit.

d In case of dosing on consecutive days, the medical assessments to be conducted on Day -1 of Period 2 may be done on Day 1 of Period 1.

e Not applicable for patients continuing from Part A.

Note: The study site should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: pharmacokinetic samples including blood sampling for blood glucose and laboratory samples per protocol nominal times. Vital sign and measurements should be scheduled before but as close as possible to the PK sampling times. Catheter-site assessments can be performed after PK sampling.

# 3. Introduction

## 3.1. Study Rationale

A prandial insulin with faster-on and/or faster-off characteristics might reduce glycaemic excursions and the incidence of postprandial hypoglycaemia compared to currently available fast-acting insulin analogues. LY900014 is an ultra-rapid-acting insulin lispro formulation with increased early absorption compared to commercially available insulin lispro formulation (Humalog®; Eli Lilly and Company). LY900014 aims to mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions. This faster profile may help to overcome commonly observed difficulties in insulin therapy in paediatric patients such as increasing weight, height, and caloric needs as well as unpredictable eating behaviour of young children and peripheral resistance of puberty.

The aim of the study is to evaluate insulin lispro pharmacokinetics (PK), glucodynamics (GD), safety, and tolerability of LY900014 in comparison with Humalog following a single subcutaneous (SC) dose administration immediately prior to a standardised liquid meal in paediatric and adult patients with type 1 diabetes mellitus (T1DM). This will be the first study to evaluate LY900014 in paediatric patients with T1DM.

## 3.2. Background

The insulin analogue, insulin lispro (Humalog), has shown to be absorbed more quickly than regular human insulin (RHI) (Humalog package insert, 2015). In healthy volunteers given SC doses of insulin lispro ranging from 0.1 to 0.4 units (U)/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). The persistent delays in absorption and variability in the duration of action of currently available rapid-acting insulin analogues contribute to commonly observed difficulties of insulin therapy in youths (due to increasing weight, height, peripheral resistance of puberty, changes in caloric needs, and unpredictable eating behaviour of younger youths; Cengiz 2012). An ultra-rapid-acting prandial insulin would shift the PK/GD of insulin analogues so that they have an even faster onset to better match carbohydrate absorption and lead to improved postprandial glucose that also allows greater flexibility in the time of dosing relative to meals.

Regional evidence suggests that the incidence of T1DM in youths is increasing in the US (Lipman et al. 2013) and Europe (EURODIAB ACE Study Group 2000). Estimates of T1DM prevalence in the paediatric population in the US come from the Centers for Disease Control and Prevention-sponsored SEARCH for Diabetes in Youth (SEARCH) Study (SEARCH for Diabetes in Youth Study Group 2009), which collects diabetes diagnosis information from multiple centres throughout the US. This study estimated the 2001 prevalence of T1DM among children under 10 years of age to be 0.76 per 1000; amongst adolescents 10 to 19 years of age, the risk more than doubled to 2.28 cases per 1000. More recently, the SEARCH study estimated that in 2011 the total number of youth in the US <20 years of age with T1DM was 166,984 (Pettit et al. 2014). As with adults, intensive diabetes management through maintenance of tight glycaemic control helps to delay onset and slow the progression of complications of the disease in youths (DCCT 1993). Tight glycaemic control is achieved via intensive insulin therapy, which requires

dietary discipline, frequent blood glucose monitoring, and multiple injections of insulin (both rapid- and long-acting insulin) to mimic the natural pattern of insulin release, the combination of which represents a major burden on the patient, especially youths. Intensive insulin treatment in T1DM should be initiated early in disease progression and the improved glycaemic control may have a lasting impact on reducing the onset of diabetic complications.

LY900014 represents a new formulation that contains insulin lispro, treprostinil, citrate, and other excipients. This formulation involves the novel use of a microdose of treprostinil [redacted] as an excipient to enhance the absorption of insulin lispro by local vasodilatation rather than as an active pharmaceutical ingredient to elicit a sy[redacted]fect. Treprostinil is a prostacyclin analogue, administered either through inhalation [redacted], as an intravenous (IV) infusion, or as a continuous SC administration for the treatment of symptomatic pulmonary arterial hypertension (PAH) and has been approved in the US since 2002 and for SC administration in Germany since 2005 [redacted] In adults dosed with LY900014, systemic treprostinil was unmeasurable and no haemodynamic effects were noted with a treprostinil dose equivalent that contained in [redacted] of LY900014. Sodium citrate, an excipient that speeds up insulin absorption (at least in part by enhancing vascular permeability), is also included in the formulation to further enhance the absorption of insulin lispro. Each of the other excipients (such as magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration (FDA)'s Generally Recognized as Safe Food Additives database and in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, concentration of the excipients in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

To date, LY900014 has been administered to approximately 89 adult patients with diabetes (60 T1DM and 29 type 2 diabetes mellitus [T2DM]). The total insulin lispro exposure was similar between LY900014 and Humalog; however, LY900014 demonstrated a faster and earlier insulin lispro absorption compared to Humalog. LY900014 displayed proportional increases in insulin lispro exposure (area under the concentration versus time curve [AUC] and maximum observed drug concentration ($C_{max}$) with dose. The faster early insulin lispro absorption was maintained across the dose range studied (7.5 to 15 U). Additionally, through the use of multiple daily injections of LY900014 for up to 2 weeks in patients with either T1DM or T2DM, it was found that LY900014 was well tolerated. Data from these Phase 1b studies show there was only 1 sample with quantifiable plasma treprostinil concentrations from approximately 472 samples taken from a total of 60 patients with T1DM and T2DM following administration of LY900014 using doses up to 50 U per SC bolus injection. In addition, no quantifiable plasma treprostinil concentrations were observed with continuous subcutaneous insulin infusion (CSII) therapy during the basal infusion or after the bolus dose in 30 patients. There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events were reported, and there were no notable increases in these events in relation to any of the LY900014 formulations compared to those in relation to Humalog. There were no reported cases of severe hypoglycaemia. Additionally, there were no reported incidences of local or systemic allergic reactions.

## 3.3. Benefit/Risk Assessment

This study will not offer any direct benefits to the patients participating in the study. The data from previous adult studies in healthy subjects and patients with T1DM and T2DM have shown that LY900014 was well tolerated and the adverse drug reactions are in keeping with those reported for Humalog.

Potential risks associated with LY900014, derived from the known risks of insulin lispro (Humalog), are hypoglycaemia, hypersensitivity reactions (localised allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral oedema (Humalog package insert, 2015).

Notably, across all doses in the adult studies that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, as described in the CCI The exposures of treprostinil in LY900014 in this study are compared to those observed in the dose ranges previously explored with SC bolus administration of treprostinil and are expected to be substantially lower than those observed in the treatment of PAH.

There are limited data on the use of treprostinil in paediatric patients. Paediatric assessments have not been required for treprostinil because of its orphan drug designation; thus, the safety and effectiveness of treprostinil in paediatric patients have not been established CCI . Subcutaneous treprostinil has been evaluated in children with PAH in an observational study (n=8). Patients initially received a fixed treprostinil dose of 1.25 ng/kg/min, which was gradually increased according to general and local tolerance, reaching a median of 40 ng/kg/min (range, 37 to 60 ng/kg/min) over the duration of follow-up (6- to 18-month time period) after initiation of treprostinil therapy. Six children had symptomatic and haemodynamic response to the addition of SC treprostinil, with manageable control of local injection-site discomfort (Levy et al. 2011). The amount of treprostinil will vary based upon the LY900014 dose administered CCI and treprostinil exposure has not been detected in adults a 1b multiple daily injection (MDI) studies or with CSII therapy during the basal infusion (up to 1.6 U/h containing 16 ng/h of treprostinil) or after the bolus dose CCI On average, the paediatric population will use a lower prepranaial insulin dose, further lowering exposure to treprostinil in this population.

In preclinical safety pharmacology and toxicity studies, or clinical pharmacology studies involving LY900014 or treprostinil alone, other than known risks associated with Humalog and CCI , no additional risks were identified. Repeat SC dosing with treprostinil sodium for up to 26 weeks was well tolerated in rats and dogs. Survival was not affected in rats at doses up to 0.1 mg/kg/day and in dogs at doses up to 0.07 mg/kg/day. Clinical signs were limited to transient, dose-related flushing of the extremities secondary to the vasodilatory pharmacology of treprostinil in rats only. No evidence of injection-site reactions to treprostinil occurred either

when treprostinil was injected alone in 13- or 26-week repeat-dose toxicity studies in rats and dogs or when treprostinil was injected in combination with insulin lispro in a 2-week local tolerance study in rats. Furthermore, there was no indication of systemic target organ toxicity in either rats or dogs. No evidence of genotoxicity was demonstrated with treprostinil in a standard battery of genetic toxicity tests. There was no evidence of reproductive toxicity or teratogenicity in rats and no evidence of teratogenicity in rabbits [redacted] hus, there are no known potential risks associated with the use of small amounts of treprostinil [redacted] in the LY900014 formulation.

Additionally, local and systemic toxicity profiles of Humalog and [redacted] do not suggest the potential for additive or synergistic toxicity.

The study includes inpatient procedures during which participants will be continuously monitored. Patients will be without further oral food intake following consumption of liquid test meal to completion of blood glucose collection (approximately 300 minutes) unless required to treat hypoglycaemia, as defined by the plasma glucose (PG) level ≤70 mg/dL (3.9 mM) with symptoms or any PG ≤60 mg/dL (3.0 mM) during the inpatient period, i.e., reversed with either rapidly absorbable oral carbohydrates or IV glucose. If a patient's blood glucose concentration rises above 300 mg/dL (16.7 mM) for more than 1 hour, RHI will be administered IV. Appropriate measures will be taken to minimise the risk of hyperglycaemia (see Section 9.4.6).

More detailed information about the known and expected benefits and risks of Humalog may be found in the following: Summary of Product Characteristics (SPC) (Humalog SPC [EPAR] [WWW]); known and expected benefits and risks of treprostinil may be found in the package insert for treprostinil ([redacted]

More information about the known and expected benefits, risks, SAEs, and reasonably anticipated AEs of LY900014 is to be found in the Investigator's Brochure (IB).

# 4. Objectives and Endpoints

Table ITSA.1 shows the objectives and endpoints of the study.

**Table ITSA.1. Objectives and Endpoints**

| Objectives | Endpoints |
|---|---|
| **Primary** | |
| 1. Part A: To evaluate the PK of insulin lispro following a single SC bolus dose administered through injection of LY900014 compared to Humalog in children, adolescents, and adults with T1DM | 1. Early 50% $t_{max}$ and AUC(0-30min) |
| 2. Part B: To evaluate the PK of insulin lispro following a single SC bolus dose administered through CSII of LY900014 compared to Humalog in children, adolescents, and adults with T1DM | 2. Early 50% $t_{max}$ and AUC(0-30min) |
| **Secondary** | |
| 1. Part A and Part B: To evaluate the difference in GD response to LY900014 and Humalog administered through SC bolus injection or CSII, as assessed using the liquid MMTT in children, adolescents, and adults with T1DM | 1. ΔAUC(0-2h), ΔAUC(0-5h) |
| 2. Part A and Part B: To evaluate the tolerability of LY900014 following SC injection or CSII in children, adolescents, and adults with T1DM | 2. AEs including injection/catheter insertion site reactions/pain, hypoglycaemia, and anti-insulin lispro antibodies |

Abbreviations: AE = adverse event; AUC = area under the concentration versus time curve; AUC(0-30min) = AUC from time zero to 30 minutes; ΔAUC = area under the baseline subtracted glucose concentration versus time curve; ΔAUC(0-2h) = area under the baseline subtracted glucose concentration versus from time 0 to 2 hours postmeal; ΔAUC(0-5h) = area under the baseline subtracted glucose concentration versus from time 0 to 5 hours postmeal; CSII = continuous subcutaneous insulin infusion; early 50% $t_{max}$ = time to early half-maximal drug concentration; GD = glucodynamics; MMTT = mixed meal tolerance test; PK = pharmacokinetic(s); SC = subcutaneous; T1DM = type 1 diabetes mellitus.

# 5. Study Design

## 5.1. Overall Design

This is a Phase 1, randomised, 2-part, patient- and investigator-blind, 2-period crossover study in children (age 6 to <12 years), adolescents (age 12 to <18 years), and adults (age 18 to <65 years) with T1DM currently using a CSII pump or MDIs. The aim is to evaluate insulin lispro PK, GD, and tolerability of LY900014 in comparison with insulin lispro (reference) following a single dose SC administration immediately prior to a standardised liquid meal.

During the screening visit, patients will receive a diary to document any AEs and/or hypoglycaemic events. Transition instructions pertinent to those patients receiving MDI who will transition their basal insulin to twice daily neutral protamine Hagedorn (NPH) will be provided and reviewed during the telephone call (on or before Day –10) discussed below.

All patients who are currently using either a CSII pump or receiving MDI and are eligible to participate in the study based on the results of the screening procedures (within 28 days of Day -1 of Period 1) will be contacted via a telephone call on or before Day -10. This telephone call will confirm their eligibility and address any questions participants and parents/guardians may have about the upcoming test meal visits. Patients entering the study on CSII therapy will remain on their current basal rate and bolus settings during the lead-in period and will be instructed to bring additional pump supplies when they arrive at the study site.

Those patients receiving MDI and who will transition from their basal insulin to twice daily NPH insulin will review the site-provided instructions during this call. The investigator will ensure the patients are provided with NPH as required. Patients or their parents/guardians may call the site at any time to address questions or concerns as they arise (including, but not limited to the transition phase). In lieu of a telephone call on or before Day -10 to review insulin transition instructions, a site visit may be scheduled at the investigator's discretion.

This study comprises of 2 parts: Part A and Part B. During Part A, patients will receive a single dose of LY900014 and Humalog on 1 occasion each as SC bolus injection and in Part B a single dose of LY900014 and Humalog on 1 occasion each as SC bolus via CSII.

Patients will be randomised on Day 1 of Period 1 in Part A and Day -1 of Period 1 in Part B. For each age group, patients will be randomised to 1 of the 2 treatment sequences (first double-blind LY900014 then double-blind Humalog, and first double-blind Humalog then double-blind LY900014) in 1:1 ratio. In both parts, dosing in Period 1 and Period 2 can occur on consecutive days but not more than 22 days apart. For patients available for dosing on consecutive days, the medical assessments scheduled for Day -1 of Period 2 may be done on Day 1 of Period 1. Patients will undergo a rescreening visit if dosing visit between Part A and Part B is more than 28 days apart. Patients who participate in Part A may continue to Part B; however, patients who complete Part A but discontinue prior to Part B may be replaced by newly enrolled patients in Part B. Newly enrolled patients will be randomised to 1 of 2 treatment sequences after completing the screening per the Schedule of Activities (Section 2) and are expected to complete

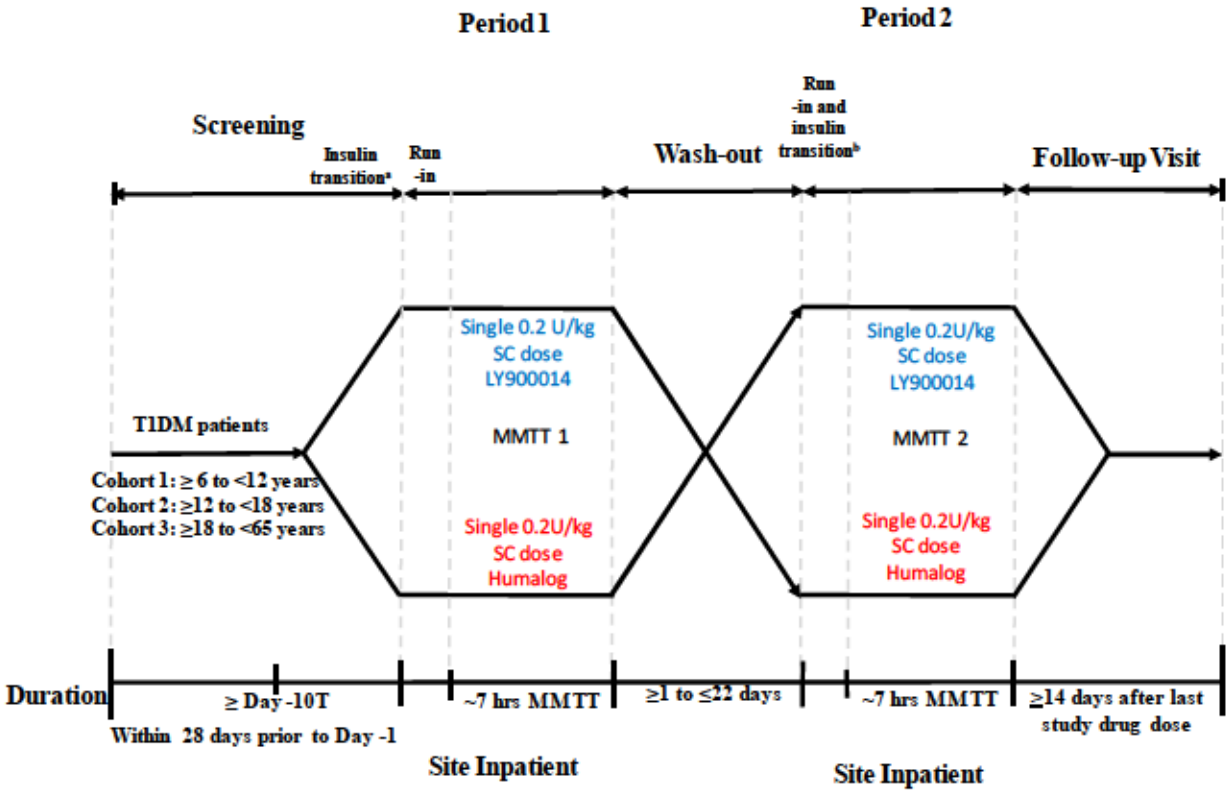
only Part B activities. Before each MMTT, a run-in period to stabilise blood glucose will occur. Details of run-in activities are described in Section 5.1.1.1, 5.1.2.1, and in Section 2.

A formal interim analysis is planned when all patients complete Part A (Section 10.3.5). Lilly intends to complete Part A to inform study design of the paediatric Phase 3 study. Initiation of Part B of the study may begin pending review of the PK, GD, safety, and tolerability data from Part A and ongoing or planned adult CSII Phase 1b studies.

### *5.1.1. Part A*

In each period, patients will undergo an overnight run-in period followed by an MMTT (see Section 5.1.1.1). Each patient will receive single 0.2 U/kg of body weight SC injections of LY900014 and Humalog. The study drug will be administered immediately before the start of a test meal (see Section 6.3.1) per the assigned treatment sequence. This dose of insulin before the MMTTs for each patient will be the same for each test meal throughout the crossover periods; this dose will also be maintained for the patient in Part B.

Figure ITSA.1 illustrates the study design for Part A.




Abbreviations: MDI = multiple daily injection; MMTT = mixed meal tolerance test; NPH = neutral protamine Hagedorn; SC = subcutaneous; T = telephone visit; T1DM = type 1 diabetes mellitus.

a At study entry, patients on MDI therapy with any basal insulin other than NPH twice daily will transition to NPH twice daily therapy approximately 7 to 3 days prior to Day -1 of Period 1.

b Insulin transition is applicable if Period 2 occurs >3 to 7 days after Period 1.

The second MMTT assessment may occur on consecutive days but no more than 22 days after the first MMTT.

**Figure ITSA.1. Illustration of study design for Part A.**

Patients will be required to attend the study site on 3 or 4 occasions for Part A as described below:

- A screening visit (up to 28 days prior to Day -1 of Period 1). The informed consent/assent will be obtained during this visit and patients will be provided a diary to document any AEs and/or hypoglycaemic events. A telephone visit is planned after screening and on or prior to Day -10 to inform patients of their eligibility, provide instructions and address any questions participants and parents may have about the upcoming test meal visits.
- Two separate inpatient overnight stays from Day –1 to Day 1 (1 in each study period for MMTTs). These inpatient stays may occur on consecutive nights.
- A follow-up visit at least 14 days after the last dose or early discontinuation if the patient discontinues after Part A.

Patients will be discharged from the study site on Day 1 (at least 7 hours after the test meal) or later, if deemed necessary for safety monitoring as determined by the investigator.

Patients entering the study on MDI therapy with any basal insulin other than NPH twice daily will transition to NPH twice daily therapy approximately 7 to 3 days prior to Day -1 of Period 1. Transition guidelines, including dosages and timing, will be determined by the investigator and will be based on the needs of individual patient.

On Day -1, the last injection of NPH prior to the MMTT will occur in the morning and will resume only after the completion of the MMTT assessment (~7 hours postdose on Day 1).

#### 5.1.1.1. Inpatient Procedures for Part A

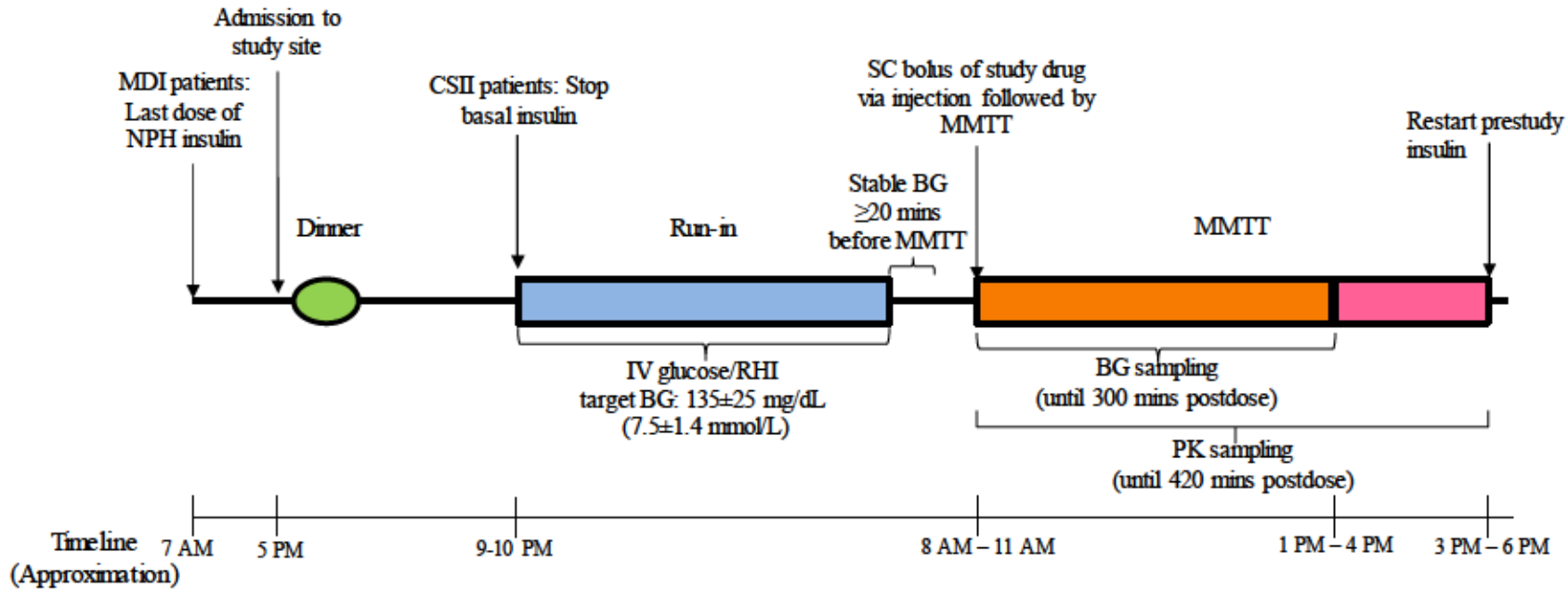
On the day before each MMTT day (i.e., Day −1 in each of the 2 periods), patients will arrive at the study site in the early evening (approximately 05:00 PM). The patient's short-acting insulin will be administered before the start of a standardised dinner. Before the start of a run-in period, the cannulation of the vein will be performed. At the start of the run-in period, patients entering the study on CSII therapy will suspend all insulin delivery and disconnect tubing from the infusion site.

Start of the run-in period will occur following the evening meal (approximately at 09:00 or 10:00 PM). The run-in period will start with a variable IV infusion of glucose (5% dextrose solution) or RHI to reach a target glucose level of 135±25 mg/dL (7.5±1.4 mmol/L). If this target glucose level is not attained before 11:00 AM on Day 1, MMTT will be halted and may be performed on a separate visit; an MMTT can only be rescheduled (up to 7 days) once for each study period. During the run-in period, blood glucose concentrations will be monitored at a minimum of 30- to 60-minute intervals (see Sections 7.4.1 and 9.6.1).

The run-in period will end once the target blood glucose level is attained and remains stable without intervention for at least 20 minutes before the scheduled start time of MMTT on Day 1. The MMTT will start in the early morning at approximately 08:00 AM with allowance up to 11:00 AM, if required, to ensure the patient's blood glucose is stable and on target, with the premeal activities as specified in the Schedule of Activities (Section 2). LY900014 or Humalog SC bolus injection will be administered immediately before the start of MMTT using an insulin syringe. The liquid test meal will be administered on Day 1 for each of the 2 treatment periods. A detailed description of the test meal is presented in Section 6.3.1. Patients are expected to complete each test meal within 15 minutes of starting the meal. Patients will be without further oral food intake following consumption of liquid test meal to completion of blood glucose collection (approximately 300 minutes) unless required to treat hypoglycaemia, as defined by the PG level ≤70 mg/dL (3.9 mM) with symptoms or any PG ≤60 mg/dL (3.0 mM) during the inpatient period, i.e., reversed with either rapidly absorbable oral carbohydrates or IV glucose. If a patient's blood glucose concentration rises above 300 mg/dL (16.7 mM) for more than 1 hour, RHI will be administered IV. In both cases, blood samples for blood glucose (for safety) will be taken and PK samples will be collected as planned. For each MMTT, the patient should stay in a sitting or semisupine position for 2 hours postdose and the patient will not be allowed to consume water for 2 hours after dosing (see Figure ITSA.2). If necessary, patients may be given

a low carbohydrate snack (<20 g carbohydrate), which does not require an insulin dose, after GD assessments (approximately 300 minutes postdose). Following completion of the assessment period and sample collection (approximately 420 minutes), patients will be offered a meal. For patients who were using MDI therapy, the investigator shall provide guidance on whether they should resume their prestudy basal insulin at the end of Period 1 or continue with NPH depending on the length of time between Period 1 and Period 2. Patients may return to pre-study basal insulin following completion of Period 2 MMTT sampling period with investigator's guidance. For CSII patients, the insulin pump may be resumed at the usual rates and settings with prestudy insulin and bolus mealtime insulin will be administered. A new infusion set may be required.

Patients will be observed following administration of the meal and discharged at the investigator's discretion.




Abbreviations: BG = blood glucose; CSII = continuous subcutaneous insulin infusion; IV = intravenous; MDI = multiple daily injection; MMTT = mixed meal tolerance test; NPH = neutral protamine Hagedorn; PK = pharmacokinetics; RHI = regular human insulin; SC = subcutaneous.

**Figure ITSA.2. Inpatient procedure in Part A.**

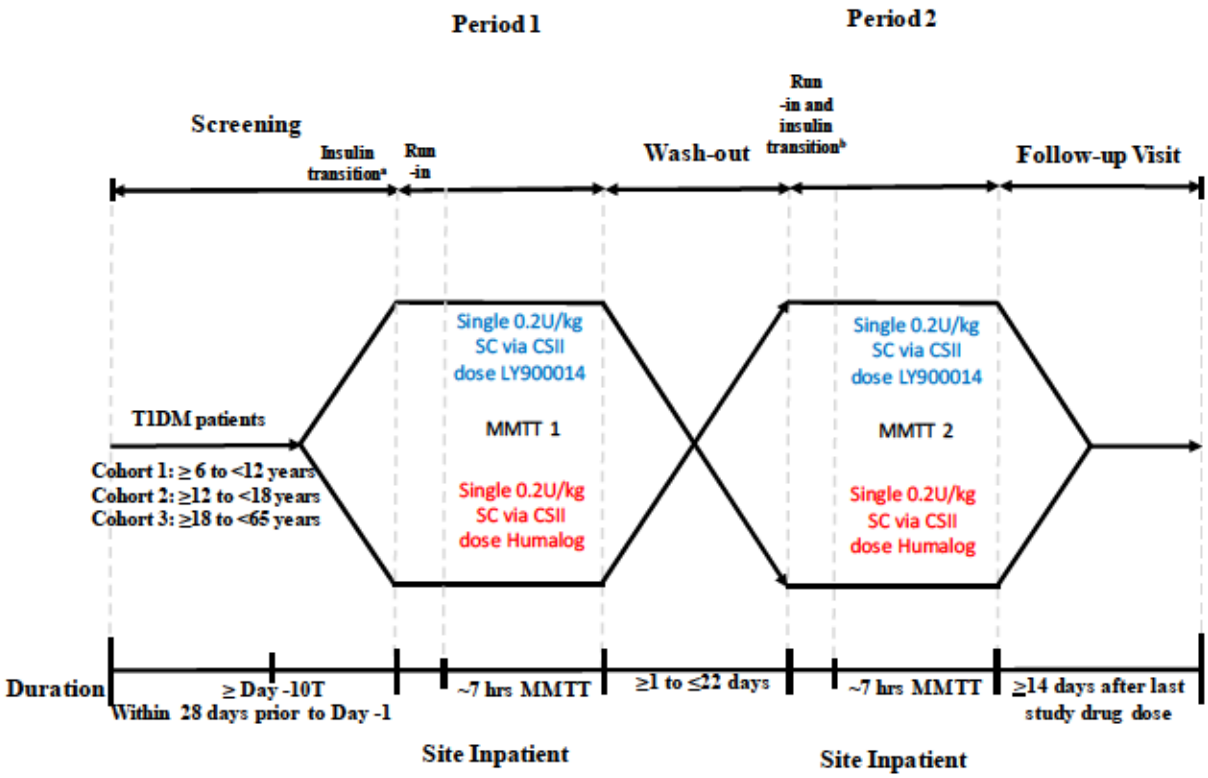
Assessment of local tolerability at all injection sites will be performed as specified in the Schedule of Activities (Section 2), including inspection of the injection/catheter site for signs such as oedema, erythema, and rash.

### *5.1.2. Part B*

Patients who completed Part A but discontinued before the beginning of Part B may be replaced by newly enrolled patients in Part B. Newly enrolled patients will be randomised to 1 of 2 treatment sequences after completing the screening per the Schedule of Activities (Section 2) and are expected to complete only Part B activities.

Patients will receive either LY900014 or Humalog with an insulin lispro dose level of 0.2 U/kg body weight, as a SC bolus dose delivered using the CSII pump per the assigned treatment sequence. Patients will receive the assigned treatment immediately before each of the test meals. At the start of dosing period (Day 1), a standardised test meal will be administered (see Section 6.3.1).

Figure ITSA.3 illustrates the study design for Part B.




Abbreviations: CSII = continuous subcutaneous insulin infusion; MDI = multiple daily injection; MMTT = mixed meal tolerance test; NPH = neutral protamine Hagedorn; SC = subcutaneous; T = telephone visit; T1DM = type 1 diabetes mellitus.

[a] Prior to Period 1, patients on MDI therapy with any basal insulin other than NPH twice daily will transition to NPH twice daily therapy approximately 7 to 3 days prior to Day -1 of Period 1.

[b] Insulin transition is applicable if Period 2 occurs >3 to 7 days after Period 1. The second MMTT assessment may occur on consecutive days but no more than 22 days after the first MMTT.

**Figure ITSA.3. Illustration of study design for Part B.**

Patients will be required to attend the study site on 3 to 4 occasions for Part B:

- A screening visit (may occur up to 28 days prior to Day -1 of Period 1). The informed consent/assent will be obtained from newly enrolled patients during this visit and a diary will be provided to document any AEs and/or hypoglycaemic events. The informed consent/assent is not applicable to patients continuing from Part A.
- A telephone visit is planned after screening and on or before Day -10 to inform patients of their eligibility and address any questions participants and parents may have about the upcoming test meal visits.
- Two separate inpatient overnight stays from Day –1 to Day 1 (1 in each study period for MMTTs). These inpatient stays may occur on consecutive nights.

- A follow-up visit at least 14 days after the last dose or early discontinuation.

Patients will be discharged from the study site on Day 1 (at least 7 hours after the test meal) or later, if deemed necessary for safety monitoring as determined by the investigator.

Patients entering the study on MDI therapy with any basal insulin other than NPH twice daily will transition to NPH twice daily therapy approximately 7 to 3 days prior to Day -1 of Period 1. On Day -1, the last injection of NPH prior to the MMTT will occur in the morning and will resume only after the completion of the MMTT assessment (~7 hours postdose on Day 1).

#### 5.1.2.1. Inpatient Procedures for Part B

On the day before each MMTT day (i.e., Day –1 in each of the 2 periods if performed on non-consecutive days), patients will arrive at the study site in the early evening (approximately 05:00 PM). Patients on CSII pump will be advised to bring a spare infusion set for the CSII pump when they arrive at the study site. Patients who are not using CCI pump will be switched to a CCI pump during the inpatient procedures. A standardised dinner will be provided where the bolus dose of the patient's usual insulin will be administered via CSII for this meal. Patients on MDI will be started on CCI pump following the evening meal. The pump reservoir will be filled with the study insulin, either LY900014 or Humalog, and a new standard infusion set and catheter will be inserted after a priming dose has been given to test the catheter. In case of immediate infusion set intolerance, a new needle can be inserted up until 12:00 AM (midnight). Following the evening meal, patients will continue their current pump basal rate until the start of the run-in period at which time a fixed low basal rate of 0.1 U/hour will be maintained until 2 hours before the scheduled start time of MMTT. For patients transitioning from MDI therapy, the bolus dose used for the evening meal will be based on their usual insulin to carbohydrate ratio and correction factor if applicable. Additionally, the basal insulin rate for the period between dinner and the start of the overnight infusion should be determined using the 80% of the pre-study total daily NPH dose divided by 24 hours to obtain the hourly basal insulin infusion rate. The new basal rate may be titrated as needed based on the investigator's discretion. Before the start of a run-in period, the cannulation of the vein will be performed. Start of the run-in period will occur following the evening meal (at approximately 09:00 or 10:00 PM). The run-in period will start with a variable IV infusion of glucose (5% dextrose solution) or RHI to reach a target glucose level of 135±25 mg/dL (7.5±1.4 mmol/L). If this target glucose level is not attained before 1100 hours on Day 1, MMTT will be halted and may be performed on a separate visit; an MMTT can only be rescheduled (up to 7 days) once for each study period. During the run-in period, blood glucose concentrations will be monitored at a minimum of 30- to 60-minute intervals.

The run-in period will end once the target blood glucose level is attained and remains stable without intervention for at least 20 minutes immediately before the scheduled start time of MMTT on Day 1. The MMTT will start in the early morning at approximately 08:00 AM with allowance up to 11:00 AM, if required, to ensure the patient's blood glucose is stable and at target, with the premeal activities as specified in the Schedule of Activities (Section 2). LY900014 or Humalog SC bolus will be administered immediately before the start of MMTT via the pump. The liquid test meal will be administered on Day 1 for each of the 2 treatment

periods. A detailed description of the test meal is presented in Section 6.3.1. Patients are expected to complete each test meal within 15 minutes of starting the meal. Patients will be without further oral food intake following consumption of liquid test meal to completion of blood glucose collection (approximately 300 minutes) unless required to treat hypoglycaemia, as defined by the PG level ≤70 mg/dL (3.9 mM) with symptoms or any PG ≤60 mg/dL (3.0 mM) during the inpatient period, i.e., reversed with either rapidly absorbable oral carbohydrates or IV glucose. If a patient's blood glucose concentration rises above 300 mg/dL (16.7 mM) for more than 1 hour, RHI will be administered IV. In both cases, blood samples for blood glucose (for safety) will be taken and PK samples will be collected as planned. For each MMTT, the patient should stay in a sitting or semisupine position for 2 hours postdose and the patient will not be allowed to consume water for 2 hours after dosing. If necessary, patients may be given a low carbohydrate snack (<20 g carbohydrate), which does not require an insulin dose, after GD assessments (approximately 300 minutes postdose). If Period 2 is performed on non-consecutive days, following completion of the assessment period and sample collection (approximately 420 minutes), the pump will be disconnected. For patients who used CSII pre-study, a new reservoir should be filled with their usual pump insulin and a new infusion set should be inserted as directed by the study site. The basal rates may be returned to the patient's usual basal settings (see Figure ITSA.4).

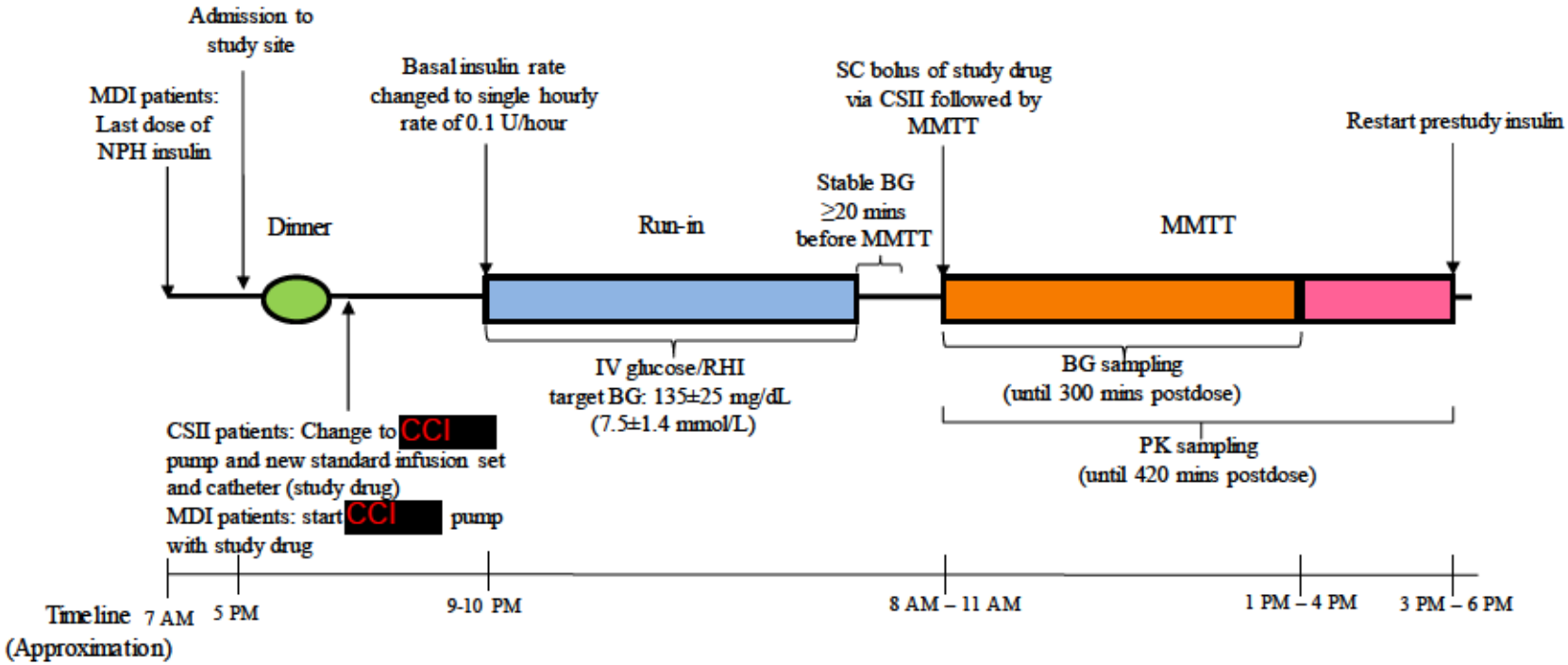
For patients who were using MDI therapy, the investigator shall provide guidance on whether they should resume their pre-study basal insulin at the end of Period 1 or continue with NPH depending on the length of time between Period 1 and Period 2.

All patients will be offered a meal at the conclusion of the MMTT assessment period.

For patients who opt to complete Period 2 on consecutive days, the pump reservoir will be filled with the other study insulin, either LY900014 or Humalog, and a new standard infusion set and catheter will be inserted prior to administration of the evening meal. In case of immediate infusion set intolerance, a new needle can be inserted up until 12:00 AM (midnight). A standardised dinner will be provided where the bolus dose for this meal will be administered via CSII. Following the evening meal, patients will continue their current pump basal rate until the start of the run-in period at which time a fixed low basal rate of 0.1 U/hour will be maintained until 2 hours before the scheduled start time of MMTT. A similar approach to that used in Period 1 will be used to manage the evening meal bolus, the run-in period, overnight basal insulin infusion, administration of study insulin bolus, and conduct of the MMTT. At the end of the MMTT assessment period, patients who used CSII pre-study should fill a new reservoir with their usual pump insulin and a new infusion set should be inserted as directed by the study site. The basal rates may be returned to the patient's usual basal settings. Patients previously treated with MDI therapy should resume their pre-study MDI therapy at the end of Period 2 as directed by the site. All patients will be offered a meal at the conclusion of the MMTT assessment period. Patients will be observed following administration of the meal and discharged at the investigator's discretion.

Assessment of local tolerability at all injection sites will be performed as specified in the Schedule of Activities (Section 2), including inspection of the injection/catheter site for signs such as oedema, erythema, and rash.

For each inpatient visit, serial blood samples will be collected over approximately 420 minutes to assess the PK and GD response following the start of each test meal.




Abbreviations: BG = blood glucose; CSII = continuous subcutaneous insulin infusion; IV = intravenous; MDI = multiple daily injection; MMTT = mixed meal tolerance test; NPH = neutral protamine Hagedorn; PK = pharmacokinetics; RHI = regular human insulin; SC = subcutaneous.

**Figure ITSA.4. Inpatient procedure in Part B.**

## 5.2. Number of Participants

Approximately 45 patients may be enrolled into the study so that approximately 36 patients (12 patients in each age group [6 to <12 years, 12 to <18 years, and 18 to <65 years] based on the age at initial screening) complete both inpatient periods in both the parts of the study.

Patients participating in Part A may continue participation in Part B. If patients from Part A decline participation in Part B or drop out from Part A, additional patients will be enrolled to ensure approximately 36 patients (12 patients in each age group) complete both periods in Part B of the study.

## 5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

## 5.4. Scientific Rationale for Study Design

Each study part consists of a 2-period crossover design to reduce the variability of insulin PK and GD, as each patient will act as his/her own control. The total number of patients needed for a crossover design is less than the number needed for a parallel group design.

Randomisation and blinding are used to avoid bias introduced through an association between allocation order of study drug and patient characteristics.

In each study period, patients will undergo a run-in period before the MMTT using a variable insulin and glucose IV infusion. This run-in will allow for improved comparability of the postprandial glucose response to a mixed meal after treatment with LY900014 and Humalog administered immediately before the start of the meal. This run-in aims to achieve similar preprandial glucose levels for all patients before the start of the test meal and thereby reduces the variability of the postprandial glucose response. Regular human insulin has been chosen for the IV optimisation of PG during the run-in because RHI does not cross-react with the insulin lispro–specific assay used for the PK analysis.

## 5.5. Justification for Dose

The bolus dose of insulin lispro (LY900014 or Humalog) will be fixed dose by body weight (0.2 U/kg). Based on previous studies of both insulin lispro (Humalog) and LY900014, a 0.2 U/kg prandial dose is necessary to provide measurable PK and GD profiles for both study insulins. This dose level has been previously examined in paediatric patients with another rapid-acting analogue with no apparent hypoglycaemic risk (Fath et al. 2017). Furthermore, the occurrence of hypoglycaemia in patients with T1DM or T2DM participating in previous test meal studies with LY900014 was also low, since patients do not receive significant basal insulin coverage during the course of the MMTT.

Meal size differs between adults and children. Due to this, the volume of the liquid test meal for the MMTT will be adjusted by weight for the child and adolescent patients within the study. However, the adult patients in the study will receive a set volume for the MMTT.

For each patient, the individualised prandial insulin lispro dose in LY900014 and Humalog for each test meal must be kept identical throughout the crossover periods for both Part A and Part B. If the investigator determines that it would be unsafe to repeat the same dose, the patient should be discontinued. A minimum weight of 25 kg is required for this study in order to have sufficient dose CCI to characterise both the serum insulin lispro absorption and elimination following study drug administration.

# 6. Study Population

Eligibility of patients for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to Day -1 of Period 1 for both Part A and Part B. Patients who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

## 6.1. Inclusion Criteria

Patients are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrolment:

[1] male or female patients with a diagnosis of T1DM based on medical history for at least 1 year as previously demonstrated by either documented presence of autoimmune antibodies (glutamic acid decarboxylases; insulin-associated tyrosine phosphatase antibody; insulin autoantibody; islet cell antibody) as per medical records, or diagnosis by an endocrinologist (paediatric or adult).

[1a] no male contraception required except in compliance with specific local government requirements

[1b] female patients:

i. Post-pubertal females are defined as children and adolescents ≥12 years of age or <12 years of age who have onset of menses

ii. Post-pubertal females of child-bearing potential – who are abstinent (if this is complete abstinence, as their preferred and usual lifestyle) or in a same sex relationship (as part of their preferred and usual lifestyle) must agree to either remain abstinent or stay in a same sex relationship without sexual relationships with males.

iii. Otherwise, post-pubertal females of child-bearing potential participating must agree to use 1 highly effective method of contraception for the entirety of the study.

1. Post-pubertal females of child-bearing potential must test negative for pregnancy prior to initiation of treatment as indicated by a negative serum pregnancy test at the screening visit followed by a negative urine pregnancy test within 24 hours prior to exposure.

2. One highly effective method of contraception. The patient may choose to use a combined (estrogen and progestogen containing) or

progestogen-only hormonal contraception administered orally, intravaginally, or transdermally and is associated with inhibition of ovulation. Alternatively, patients may use either an intrauterine device, intrauterine hormone-releasing system, bilateral tubal occlusion, or the partner should have been vasectomised.

iv. post-pubertal females not of child-bearing potential may participate and include those who are

1. infertile due to surgical sterilisation (hysterectomy or bilateral oophorectomy or bilateral salpingectomy), congenital anomaly such as Mullerian agenesis; or

2. post-menopausal – defined as no menses for 12 months without an alternative medical cause. A follicle-stimulating hormone level in the post-menopausal range may be used to confirm a post-menopausal state in women not using hormonal contraception or hormonal replacement therapy.

[2] are aged ≥6 and <65 years

[3] children and adolescents have a body mass index (BMI) within the 3rd and 95th BMI percentiles with a minimum weight of 25 kg; adults have a BMI ≤28.0 kg/m$^2$

[4] continuous treatment with insulin for at least 12 months and with at least 6 months of treatment using CSII pump therapy or MDI therapy consisting of treatment with basal insulin (including insulin glargine U100, detemir, degludec U100 and NPH) and rapid-acting insulin prior to the main meals of the day before screening

[5] total daily dose of insulin between 0.3 and 1.5 U/kg/day, inclusive

[6] have a glycated haemoglobin (HbA1c) ≤10% at screening

[7] compliant with prescribed diet and insulin therapy, as determined by the investigator

[8] have clinical laboratory test results within normal reference range (with the exception of HbA1c) for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator

[9] have venous access sufficient to allow for glucose infusion and blood sampling procedures as per protocol

[10] are reliable and willing to make themselves/their wards available for the duration of the study and are willing to follow study procedures

[11] are able and willing to give signed informed consent approved by Lilly and the ethical review board (ERB) governing the site or both the child/adolescent and a parent or legal guardian are able to understand and fully participate in the activities of the clinical trial and sign their assent and consent, respectively

## 6.2. Exclusion Criteria

Patients will be excluded from study enrolment if they meet any of the following criteria at screening and/or enrolment:

[12] receiving any oral or injectable medication intended for the treatment of diabetes mellitus other than insulins in the 12 months prior to screening

[13] more than 1 episode of severe hypoglycaemia (defined as requiring assistance due to neurologically disabling hypoglycaemia, indicated by coma or convulsion and/or use of IV glucose or glucagon) within 6 months prior to screening

[14] two or more emergency room visits or hospitalisations due to poor glucose control (hyperglycaemia or diabetic ketoacidosis) within 6 months of screening

[15] post-pubertal females not using a reliable method of birth control, or intending to become pregnant

[16] one of the following concomitant diseases: presence of clinically significant hematologic, oncologic, renal, cardiac, hepatic, or gastrointestinal disease, proliferative retinopathy, uncontrolled celiac disease, uncontrolled hyperthyroidism or hypothyroidism, or adrenal insufficiency

[17] have a history of renal impairment (exclusion only if glomerular filtration rate [estimated GFR] <60 mL/minute/1.73 $m^2$ [GFR is estimated according to the Chronic Kidney Disease Epidemiology Collaboration creatinine equation]) (Schwartz et al. 2009).

[18] hepatic: Have obvious clinical signs or symptoms of liver disease (e.g., acute or chronic hepatitis as indicated by evidence and/or positive antibody/surface antigen to hepatitis B/C, or cirrhosis) or elevated liver enzyme measurements as indicated below at screening:

a. total bilirubin level (TBL) ≥2× the upper limit of normal (ULN) (with the exception of Gilberts disease) as defined by the central laboratory,

   OR

b. alanine aminotransferase (ALT) ≥3× ULN as defined by the central laboratory

   OR

c. aspartate aminotransferase (AST) ≥3× ULN as defined by the central laboratory

[19] are study site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling

[20] are Lilly employees or are employees of the study site and their children

[21] are currently enrolled in a clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study

[22] have participated, within the past 30 days, in a clinical study involving an investigational product. If the previous investigational product has a long half-life ($t_{1/2}$), 5 half-lives or 30 days (whichever is longer) should have passed

[23] have previously completed or withdrawn from this study

[24] have known allergies to treprostinil CCI insulin lispro, RHI, related compounds, or any components of the formulation, or a history of significant atopy

[25] Adults ≥18 years of age who have donated blood of more than 450 mL or more in the past 3 months before screening or

any prospective study participant who has experienced severe blood loss within 3 months prior to screening or have known haemoglobinopathy, haemolytic anaemia, or sickle cell anaemia, or any other traits of haemoglobin abnormalities known to interfere with the HbA1c measurement

[26] receiving chronic (lasting longer than 14 consecutive days) systemic glucocorticoid therapy (excluding topical, intraocular, intranasal, and inhaled preparations) or have received such therapy within 8 weeks immediately before screening

[27] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator

[28] have a history of in-patient psychiatric treatment, emotional, behavioural, or other untreated conditions that would interfere with proper participation in routine diabetes control and management in the past 6 months

[29] are unwilling to comply with the dietary requirements/restrictions during the study: (i) comply with the fasting requirements of the study, (ii) consume only the meals/snacks provided during the inpatient visits

[30] have, except for current regimen of insulin therapy and concomitant medication(s) (e.g., antihypertensive medication, lipid-lowering agent, thyroid hormone replacement medication, hormonal contraception, hormonal replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that treat hyperglycaemia or insulin resistance or that promote weight loss within 14 days before dosing

[31] regularly use known drugs of abuse and/or show positive findings on drug screening

[32] show evidence of human immunodeficiency virus (HIV) infection and/or positive human HIV antibodies

[33] are women who are lactating

[34] have an average weekly alcohol intake that exceeds 21 units per week (males) and 14 units per week (females), or are unwilling to stop alcohol consumption for the duration of the study
(1 unit = 12 oz or 360 mL of beer; 5 oz or 150 mL of wine; 1.5 oz or 45 mL of distilled spirits)

[35] are unable to comply with smoking restrictions during the study

[36] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study

### 6.2.1. Additional Exclusion Criteria for Inpatient Dosing Days

[37] consumption of alcohol within 24 hours prior to admission

[38] positive serum pregnancy test at screening or positive urine pregnancy test at admission to the study site

[39] consumption of coffee, tea, chocolate, cola, and/or energy drinks containing methyl xanthine (caffeine, theophylline, or theobromine) within 12 hours before each test meal assessment.

[40] strenuous exercise within the past 24 hours before each test meal assessment

[41] any medical condition or AE that could interfere with glucose metabolism, as judged by the investigator

[42] any use of excluded prescription or nonprescription medication from 14 days prior to dosing and throughout the study

[43] hypoglycaemia during the MMTT period and less than 24 hours before dosing that poses a significant risk to patient's safety, as judged by the investigator

[44] injection of a bolus of more than 6 U of a fast-acting insulin analogue 6 hours before dosing prior to the test meal

### 6.2.2. Rationale for Exclusion of Certain Study Candidates

The blood volume to be withdrawn per patient in each part of this study is approximately 170 mL which would be a challenge in children <6 years of age. Additionally, insulin is individually titrated and therefore, the understanding of PK and GD in the proposed patients (≥6 years to ≤64 years) should be sufficient to inform dosing in children <6 years of age under the supervision of a qualified endocrinologist. Therefore, this study will not include paediatric patients with T1DM from birth to <6 years of age.

## 6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients may undergo medical assessments and review of compliance with requirements before continuing in the study.

### *6.3.1. Meals and Dietary Restrictions*

Patients will be provided standardised liquid meals for the MMTTs, as outlined in the Schedule of Activities (Section 2). An evening meal on the day prior to the MMTT and a meal following completion of the MMTT will also be provided. Patients will be fasted (except for water) for at least 10 hours before each test meal and consume each meal within approximately 15 minutes. The standardised liquid test meal will use CCI For children and adolescents in the study, the volume of the liquid test me ed on the patient's body weight (kg) during the study (Period 1 of Part A or Period 1 of Part B if patients did not participate in Part A of the study) as described in Table ITSA.2. For adults, the liquid test meal will consist of CCI approximately 100 g of carbohydrates total in test meal). pt consistent with regard to calorie and nutrient content across all MMTT assessments in the study. For both parts of the study, the patient will not be allowed to consume water for 2 hours after dosing apart from fluid provided with the meal; however, water may be consumed freely afterwards.

**Table ITSA.2. Volume of Standardised Meal According to Body Weight**

| Body Weight (kg) | Volume of Meal (mL) |
|---|---|
| 25-29 | 230 |
| 30-34 | 272 |
| 35-39 | 315 |
| 40-44 | 357 |
| 45-49 | 400 |
| 50-54 | 442 |
| >55 | 474CCI |

While resident in the study site, patients may not consume any food or caloric drinks other than that provided by the study site. When not resident in the study site, patients may resume their regular diet.

Following completion of study procedures, patient will be offered a meal. The study site will record dose and time of prandial insulin administered with end-of-study meal.

### *6.3.2. Caffeine, Alcohol, and Tobacco*

Patients should refrain from caffeine-containing food/beverages (e.g., cola, chocolate drinks, tea, coffee, energy drinks containing methylxanthine [caffeine, theophylline, or theobromine]) for at least 12 hours before each test meal and throughout the duration of each study site visit, excluding the test meal when CCI may be given.

No alcohol will be allowed at least 24 hours before each study site admission (Day –1) and throughout the duration of each study site visit. Between study site visits, daily alcohol consumption should not exceed 21 units per week for males and 14 units per week for females.

No cigarette smoking will be permitted during the inpatient visits.

### *6.3.3. Activity*

Patients will be encouraged to maintain their regular exercise and insulin regimen adaptation related to exercise during the outpatient period; however, they should not undertake vigorous or prolonged exercise at least 24 hours before each dosing day at the study site. Movement will be restricted to retain the integrity of connections to infusion(s) and the study procedures.

## 6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may be re-screened once. The interval between re-screenings should be at least 28 days. Each time re-screening is performed, a new informed consent or assent form (see Appendix 3 for details) must be signed and participants will be assigned a new identification number.